

REV2021

SEMESTER 2

3 CREDITS

Textile Fibres

Course Code: 2061 | Program Core

[Official Syllabus Source](#)

Course Overview

Teaching Scheme

- Lectures: 2 Hours/Week
- Tutorials: 1 Hour/Week
- Total Periods: 45 Hours

Assessment Scheme

- Continuous Internal Evaluation (CIE)
- End Semester Examination (ESE)
- Two Series Tests

This course provides foundational knowledge on natural vegetable, animal, and semi-synthetic fibres. It covers their chemical structure, physical properties, and industrial applications, with a specific focus on cotton, silk, wool, and synthetic polyamides like Nylon 6 and 6,6.

Course Outcomes (COs)

CO No.	Description	Cognitive Level
CO1	Identify classification and general properties of textile fibres. Illustrate structure, properties & uses of Cotton.	Understanding
CO2	Describe structure, properties and uses of Bast fibres (Linen, Jute) & Animal fibres (Silk, Wool).	Understanding
CO3	Explain manufacturing process, structure, properties and uses of Semi-synthetic fibres.	Understanding
CO4	Explain types of polymerizations and manufacturing process, properties and uses of Nylon 6.6 & Nylon 6.	Understanding

Module 1 | 11 Hours

Classification & Cotton Fibre

1.1 Classification of Textile Fibres—

Textile fibres are the fundamental units of textile structures. They are classified based on their **Origin** and **Chemical Nature**.

Classification by Origin

- **Natural Fibres:**
 - Vegetable (Cellulosic): Cotton, Linen, Jute, Hemp.
 - Animal (Protein): Silk, Wool.
 - Mineral: Asbestos.
- **Man-made Fibres:**
 - Regenerated (Semi-synthetic): Viscose Rayon, Acetate Rayon.
 - Synthetic: Nylon, Polyester, Acrylic.
 - Inorganic: Glass, Metallic fibres.

Malayalam Note: ടെക്സൈൽ ഫൈബറുകൾ ടെക്സൈൽ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ (Fibre). ടെക്സൈൽ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ (Natural) ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ (Man-made) ഓർഗനൈസേഷൻ ഓർഗനൈസേഷൻ. ഓർഗനൈസേഷൻ (Cotton),

रेशम (Silk), ऊन (Wool) प्राकृतिक रेशमयुक्त हैं। रेशम, ऊन प्राकृतिक रेशमयुक्त हैं।

1.2 Cotton: Structure & Properties—

Cotton is a natural vegetable fibre obtained from the seed hair of the *Gossypium* plant.

Structure of Cotton

Longitudinal View (Convolutions)



Cross Section (Kidney

Fig: Longitudinal view showing natural twists (convolutions) and kidney-shaped cross-section.

Physical Properties

- **Length:** 1.25 cm to 6 cm (Staple fibre).
- **Strength:** Increases when wet (unique property).
- **Fineness:** Varies by variety (Sea Island is finest).
- **Colour:** Usually white to creamy.

Chemical Properties

- **Effect of Acid:** Disintegrated by hot dilute or cold concentrated mineral acids.

- **Effect of Alkali:** Resistant to alkalis; swelling occurs in strong NaOH (Mercerization).
- **Microorganisms:** Attacked by mildew and fungi in damp conditions.

Module 2 | 11 Hours

Bast & Animal Fibres

2.1 Bast Fibres: Linen and Jute—

Bast fibres are obtained from the stem (phloem) of plants.

Linen (Flax)

Extraction: Processed via *Retting* (bacterial action in water to loosen fibres from the stem) followed by Scutching and Hackling.

Properties: Stronger than cotton, high luster, good heat conductor (cool to wear).

Jute

Known as the "Golden Fibre". It is cheap, strong, but coarse and harsh. Used mainly for packaging (gunny bags) and carpet backing.

2.2 Animal Fibres: Silk and Wool—

Silk (The Queen of Fibres)

Sericulture: The commercial rearing of silkworms (*Bombyx mori*) for the production of silk.

Lifecycle: Egg → Larva (Caterpillar) → Pupa (in Cocoon) → Moth.

Degumming: Removing the silk gum (Sericin) using soap and water to reveal the lustrous fibroin filaments.

Wool

Scouring: The process of washing raw wool to remove grease (lanolin), dirt, and dried sweat (suint).

Carbonisation: Treatment with acid to remove vegetable matter (burrs, seeds) from wool.

Felting: The interlocking of wool scales under heat, moisture, and pressure.

Module 3 | 10 Hours

Semi-Synthetic Fibres

3.1 Viscose Rayon—

Viscose Rayon is regenerated cellulose. It is made from wood pulp or cotton linters.

Manufacturing Flowchart

Wood Pulp → Steeping (NaOH) → Shredding (Alkali Cellulose) → Ageing → Xanthation (CS₂) → Dissolving → Ripening → Spinning (Wet Spinning) → Regeneration.

Properties

- Highly absorbent.
- Loses 50% strength when wet.
- Good drape and soft feel.

3.2 Modern Semi-Synthetics—

- **Lyocell:** Produced using a solvent spinning process (NMMO), more eco-friendly than viscose.
- **Modal:** A type of rayon that is highly resistant to shrinkage and fading.
- **Bamboo Fibre:** Regenerated cellulose from bamboo pulp, known for antibacterial properties.

Module 4 | 11 Hours

Synthetic Fibres & Polymerization

4.1 Principles of Polymerization—

Monomer: The small repeating unit.

Polymer: A long chain molecule made of many monomers.

Types of Polymerization

1. **Addition Polymerization:** Monomers join without loss of any small molecules (e.g., Polyethylene).
2. **Condensation Polymerization:** Monomers join with the elimination of small molecules like water or ammonia (e.g., Nylon 6,6).

4.2 Polyamides: Nylon 6,6 and Nylon 6—

Nylon 6,6

Made from **Hexamethylene diamine** and **Adipic acid**. Both monomers have 6 carbon atoms, hence the name "6,6".

Nylon 6

Made from a single monomer: **Caprolactam**. It involves ring-opening polymerization.

Spinning Methods

Method	Mechanism	Example
Melt Spinning	Polymer is melted and extruded into cold air.	Nylon, Polyester
Dry Spinning	Polymer dissolved in solvent; solvent evaporated by hot air.	Acetate, Acrylic
Wet Spinning	Polymer solution extruded into a chemical coagulating bath.	Viscose Rayon

Formula Bank & Calculations

Draw Ratio Calculation

Drawing is the process of stretching the extruded filament to align polymer chains and increase strength.

Formula: $\text{Draw Ratio} = L_{\text{final}} / L_{\text{initial}}$

Example: A nylon filament of length 10cm is stretched to 45cm. Calculate the draw ratio.

Given: $L_{\text{initial}} = 10\text{cm}$, $L_{\text{final}} = 45\text{cm}$

Calculation: $45 / 10 = 4.5$

Answer: Draw Ratio is 4.5:1.

Interactive Draw Ratio Calculator

Initial Length (m):

e.g. 1

Final Length (m):

e.g. 4

Calculate Ratio

Diagram Library



Linear Polymer (e.g. Nylon)



Linear Polymer Structure



Melting



Extrusion



Melt Spinning Schematic

Module-wise Practice Questions

1. Define Textile Fibre. (2 Marks)

A textile fibre is a unit of matter characterized by flexibility, fineness, and a high ratio of length to thickness, which renders it capable of being spun into yarn.

2. Differentiate between Nylon 6 and Nylon 6,6. (5 Marks)

- **Nylon 6:** Produced from Caprolactam; contains 6 carbons in the repeating unit; lower melting point ($\sim 215^{\circ}\text{C}$).

- **Nylon 6,6:** Produced from Hexamethylene diamine and Adipic acid; contains two 6-carbon units; higher melting point ($\sim 265^{\circ}\text{C}$).

3. Explain the process of 'Retting' in Linen production. (3 Marks)

Retting is the process of using moisture and bacterial action to dissolve the cellular tissues and pectins that bind flax fibres to the stem. Types include Water retting, Dew retting, and Chemical retting.

Practice Model Paper

Time: 3 Hours | Max Marks: 75

Part A (Answer all questions - $10 \times 2 = 20$ Marks)

1. What is a staple fibre?
2. Define Sericulture.
3. What is the chemical composition of cotton?
4. Name two bast fibres.
5. What is the monomer of Nylon 6?
6. Define Polymerization.
7. What is Scouring?
8. List two uses of Jute.
9. What is the function of a spinneret?
10. Name a solvent used in dry spinning.

Part B (Answer any 5 - $5 \times 5 = 25$ Marks)

1. Explain the classification of fibres with a chart.
2. Describe the physical properties of Silk.
3. Explain the manufacturing of Viscose Rayon.
4. Compare Melt spinning and Wet spinning.
5. Explain the effect of acids and alkalis on Cotton.

Part C (Answer any 3 - $3 \times 10 = 30$ Marks)

1. Explain the structure, properties, and uses of Wool in detail.
2. Describe the manufacturing process of Nylon 6,6 with a flowchart.

3. Discuss the extraction and properties of Linen.

Quick Revision

Fibre	Type	Key Feature
Cotton	Natural Vegetable	Convolutions, Kidney cross-section.
Silk	Natural Animal	Continuous filament, Sericin gum.
Wool	Natural Animal	Surface scales, Felting property.
Viscose	Regenerated Cellulose	High absorbency, low wet strength.
Nylon	Synthetic Polyamide	High strength, thermoplastic.

Glossary

Convolution:

The natural twist found in cotton fibres.

Lumen:

The central hollow canal in a cotton fibre.

Thermoplastic:

A material that softens when heated and hardens when cooled.

Spinneret:

A metal plate with tiny holes through which polymer dope is extruded to form filaments.

Official Source Declaration

This content is based on the **SITTTR Kerala Revision 2021** syllabus for Course 2061 (Textile Fibres). The structure, module distribution, and learning outcomes are derived directly from the official document.

References:

- V.A. Shenai, Textile Fibres - Vol. 1, 2
- S.P. Mishra, Fibre Science and Technology
- J. Gordon Cook, Handbook of Textile Fibres

മലയാളം പിന്തുണ (Malayalam Support)

ഈ അദ്ധ്യായം പൂർണ്ണമായും മലയാളത്തിൽ എഴുതിയിരിക്കുന്നു. ഇതിൽ ഉൾക്കൊള്ളുന്ന വിവരങ്ങൾ താഴെ പറയുന്നവയാണ്:

- Module 1:** കോട്ടൺ (Cotton) നൂൽപ്പണി.
- Module 2:** ജൂട്ട (Jute) നൂൽപ്പണി, ജൂട്ട നൂൽപ്പണി, ജൂട്ട നൂൽപ്പണി.
- Module 3:** ജൂട്ട നൂൽപ്പണി, ജൂട്ട നൂൽപ്പണി.
- Module 4:** ജൂട്ട നൂൽപ്പണി, ജൂട്ട നൂൽപ്പണി.

ഈ അദ്ധ്യായം പൂർണ്ണമായും മലയാളത്തിൽ എഴുതിയിരിക്കുന്നു. ഇതിൽ ഉൾക്കൊള്ളുന്ന വിവരങ്ങൾ താഴെ പറയുന്നവയാണ്: (Manufacturing), (Properties), (Uses) എന്നിവ.

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